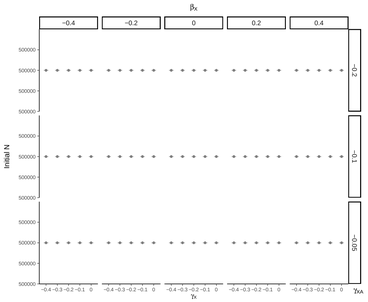
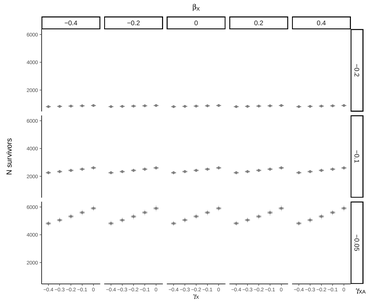
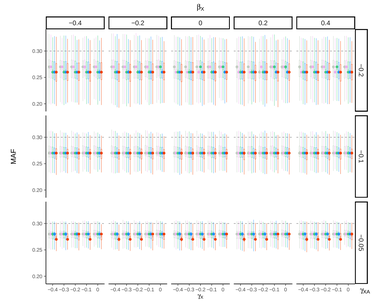
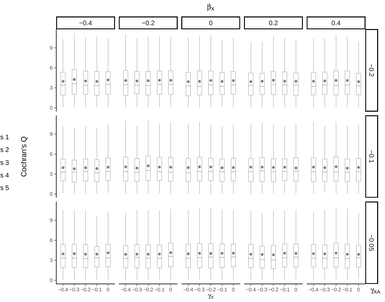
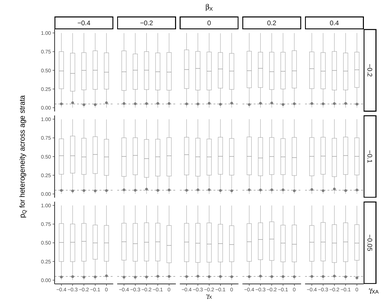
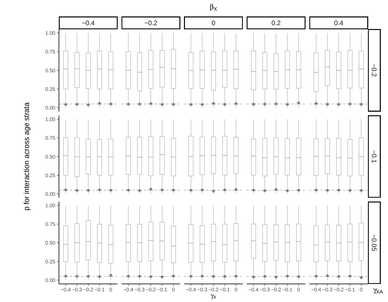

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
Initial N
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000
5500000

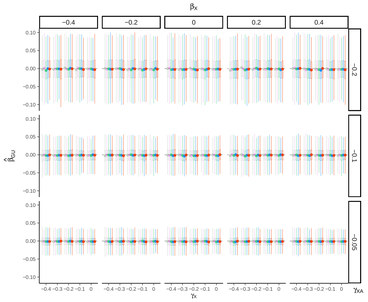

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
N survivors
6000
4000
2000
6000
4000
2000
6000
4000
2000

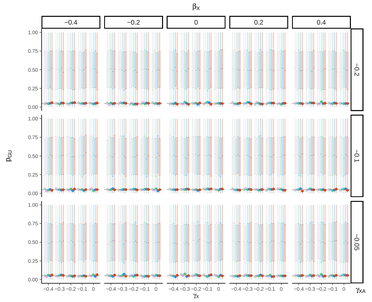

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
MAF
0.30
0.25
0.20
0.30
0.25
0.20
0.30
0.25
0.20
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

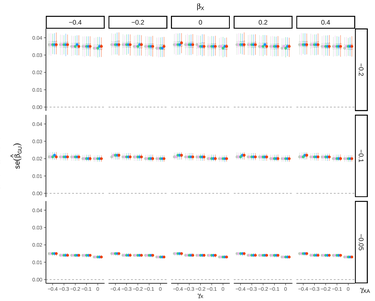

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
Cochran's Q
9
6
3
0
9
6
3
0
9
6
3
0

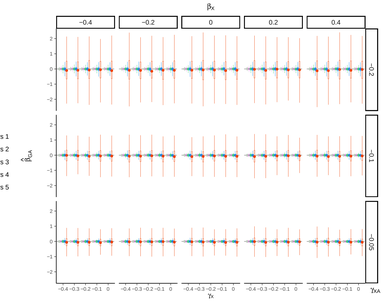

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
p0 for heterogeneity across age strata
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00

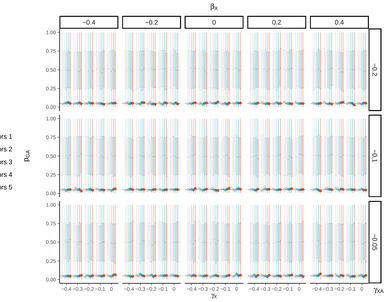

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
p for interaction across age strata
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00

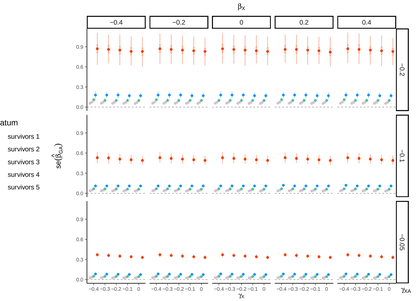

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xU
0.10
0.05
0.00
−0.05
−0.10
0.10
0.05
0.00
−0.05
−0.10
0.10
0.05
0.00
−0.05
−0.10
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

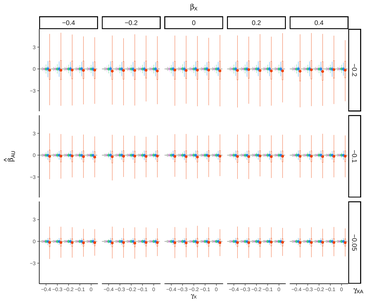

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxU
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

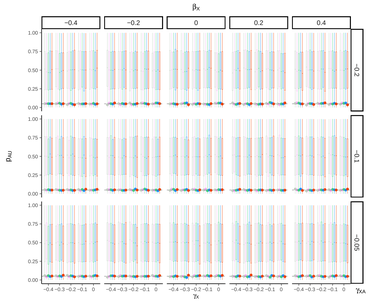

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xU)
0.04
0.03
0.02
0.01
0.00
0.04
0.03
0.02
0.01
0.00
0.04
0.03
0.02
0.01
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

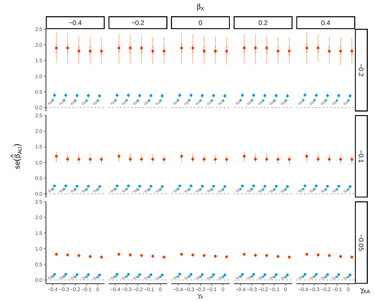

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xA
2
1
0
−1
−2
2
1
0
−1
−2
2
1
0
−1
−2
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

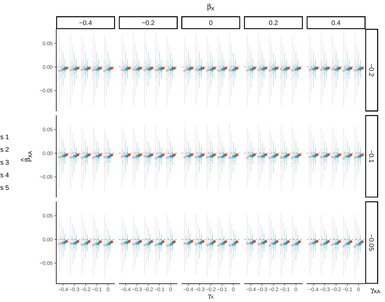

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxA
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

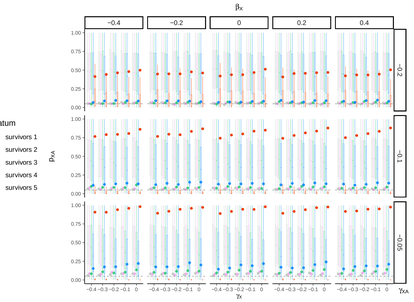

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xA)
0.9
0.6
0.3
0.0
0.9
0.6
0.3
0.0
0.9
0.6
0.3
0.0
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

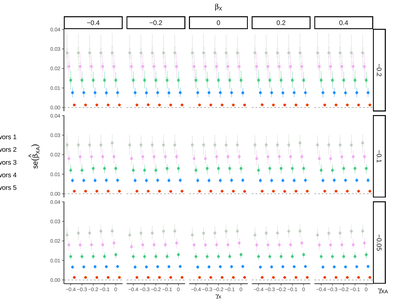

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xU
3
0
−3
3
0
−3
3
0
−3
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

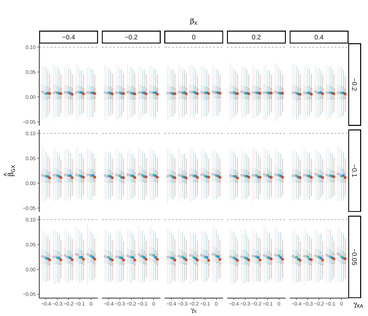

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxU
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

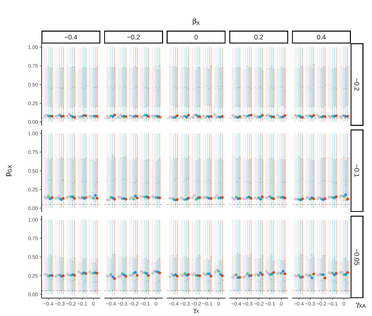

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xU)
2.5
2.0
1.5
1.0
0.5
0.0
2.5
2.0
1.5
1.0
0.5
0.0
2.5
2.0
1.5
1.0
0.5
0.0
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

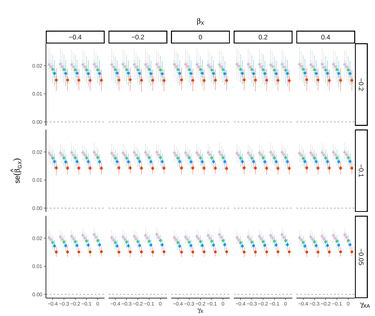

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xA
0.05
0.00
−0.05
0.05
0.00
−0.05
0.05
0.00
−0.05
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

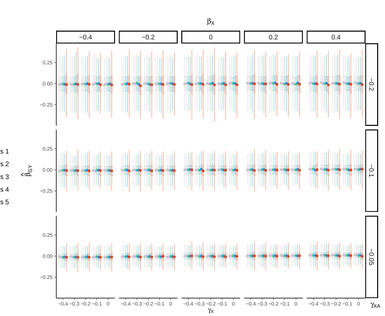

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxA
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

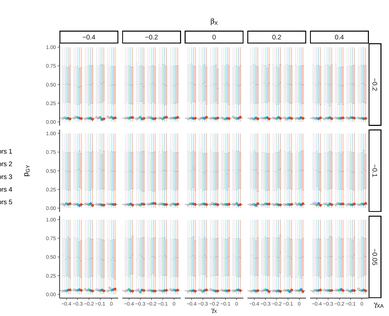

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xA)
0.04
0.03
0.02
0.01
0.00
0.04
0.03
0.02
0.01
0.00
0.04
0.03
0.02
0.01
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

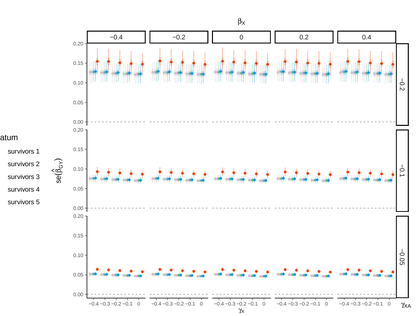

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xH
0.10
0.05
0.00
−0.05
0.10
0.05
0.00
−0.05
0.10
0.05
0.00
−0.05
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

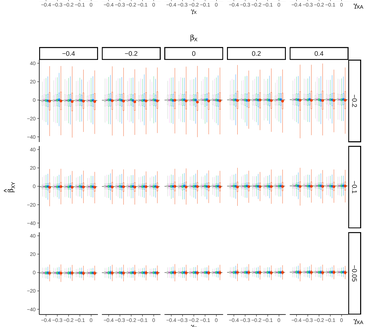

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxH
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

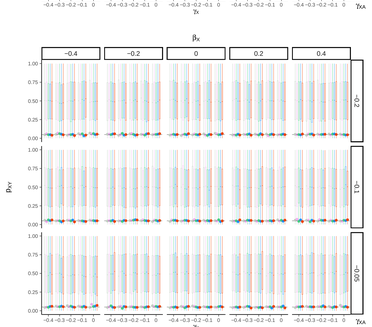

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xH)
0.02
0.01
0.00
0.02
0.01
0.00
0.02
0.01
0.00
0.02
0.01
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5

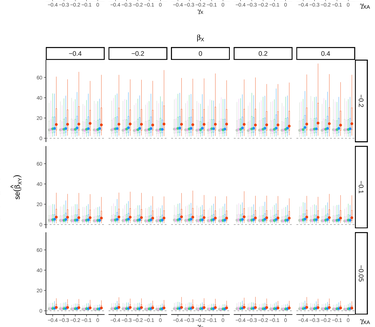

βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xT
0.25
0.00
−0.25
0.25
0.00
−0.25
0.25
0.00
−0.25
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5


βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxT
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5


βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xT)
0.20
0.15
0.10
0.05
0.00
0.20
0.15
0.10
0.05
0.00
0.20
0.15
0.10
0.05
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5


βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
β̂xY
40
20
0
−20
−40
40
20
0
−20
−40
40
20
0
−20
−40
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5


βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
pxY
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5


βX
−0.4
−0.2
0
0.2
0.4
γXA
−0.2
−0.1
−0.05
Tx
se(β̂xY)
60
40
20
0
60
40
20
0
60
40
20
0
stratum
survivors 1
survivors 2
survivors 3
survivors 4
survivors 5